

**THE UNIVERSITY OF ILLINOIS AT CHICAGO**  
**Department of Chemical Engineering**

**Homework Set #8**

**Team Homework**

**Course:** CHE 301. CHEMICAL ENGINEERING THERMODYNAMICS  
**Term:** Fall 2018  
**Professor:** Dr. Michael Caracotsios  
**E-mail:** [mcaracot@uic.edu](mailto:mcaracot@uic.edu)

**Your names:** Milos Markovic & Caleb Tristano  
**Due Date:** Friday December 04, 2018 in my office by 4:30pm

**Number of points.....100**

**Your homework score.....76**

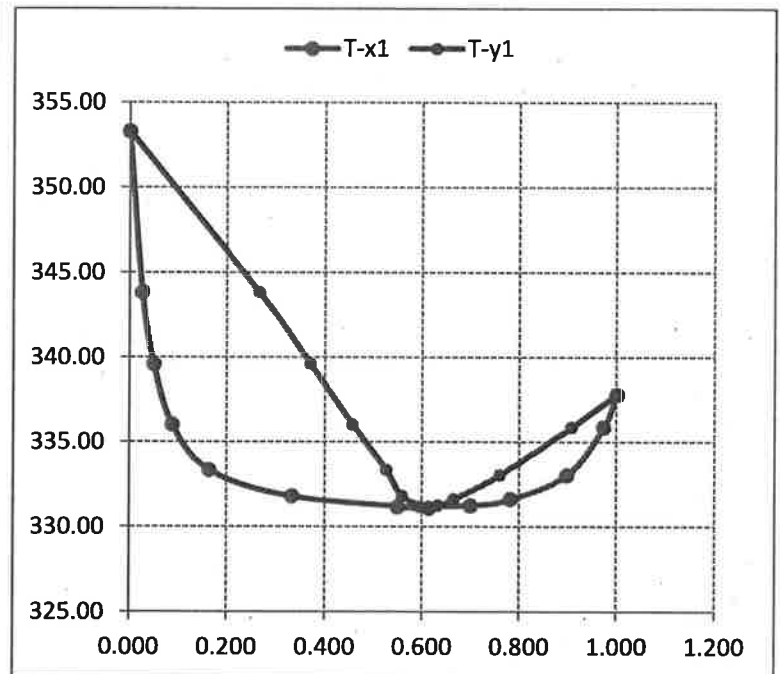
**Note:** All problems are worth 100 points. Your final score will be the average value

### Problem 1: VLE Calculations with Equations of State

Consider a mixture of Methanol (Component #1) and Benzene (Component #2) at 1.01325 bar that follows the Peng-Robinson Equation of State.

1. Assume that the binary interaction parameter  $k_{12}$  is zero. Calculate the bubble point temperature at values of  $x_1$  shown below and the dew point temperature at values of  $y_1$  shown below. Plot the bubble point temperatures vs. liquid mole fraction  $x_1$  and dew point temperature vs. vapor mole fraction  $y_1$ . Superimpose the experimental data given below onto your graph and comment on the goodness of fit, by calculating the mean residual sum of squares. A value less than 10.0 indicates an acceptable result.
2. At the azeotrope, the composition of liquid and vapor phases are the same at equilibrium. Use trial and error to determine the temperature and composition that is predicted for the azeotrope between methanol and benzene at 1.01325 bar if  $k_{12}$  is zero. Compare your result with the actual values given in the data below.
3. Determine the optimal value of the binary interaction parameter  $k_{12}$  that can be used for the prediction of the azeotrope of this mixture

| Temp(C) | $x_1$ | $x_2$ | $y_1$ | $y_2$ |
|---------|-------|-------|-------|-------|
| 80.15   | 0.000 | 1.000 | 0.000 | 1.000 |
| 70.67   | 0.026 | 0.974 | 0.267 | 0.733 |
| 66.44   | 0.050 | 0.950 | 0.371 | 0.629 |
| 62.87   | 0.088 | 0.912 | 0.457 | 0.543 |
| 60.20   | 0.164 | 0.836 | 0.526 | 0.474 |
| 58.64   | 0.333 | 0.667 | 0.559 | 0.441 |
| 57.95   | 0.614 | 0.386 | 0.614 | 0.386 |
| 58.02   | 0.549 | 0.451 | 0.595 | 0.405 |
| 58.10   | 0.699 | 0.301 | 0.633 | 0.367 |
| 58.47   | 0.782 | 0.218 | 0.665 | 0.335 |
| 59.90   | 0.898 | 0.102 | 0.760 | 0.240 |
| 62.71   | 0.973 | 0.027 | 0.907 | 0.093 |
| 64.65   | 1.000 | 0.000 | 1.000 | 0.000 |



### **Problem 2: VLE Calculations with Liquid Activity Coefficients**

The following  $P_{xy}$  data for a Toluene/n-Decane mixture are available at  $T = 373.47^\circ\text{K}$  :

| TEMPERATURE | PRESSURE | $X_1$   | $X_2$    | $Y_1$   | $Y_2$    |
|-------------|----------|---------|----------|---------|----------|
| K           | Pa       | TOLUENE | n-DECANE | TOLUENE | n-DECANE |
| 373.47      | 9450     | 0.0000  | 1.0000   | 0.0000  | 1.0000   |
| 373.47      | 11630    | 0.0353  | 0.9647   | 0.2415  | 0.7585   |
| 373.47      | 14180    | 0.0700  | 0.9300   | 0.3834  | 0.6166   |
| 373.47      | 17240    | 0.1223  | 0.8777   | 0.5147  | 0.4853   |
| 373.47      | 22710    | 0.1880  | 0.8120   | 0.6544  | 0.3456   |
| 373.47      | 29530    | 0.3006  | 0.6994   | 0.7732  | 0.2268   |
| 373.47      | 34260    | 0.3665  | 0.6335   | 0.8277  | 0.1723   |
| 373.47      | 45240    | 0.5301  | 0.4699   | 0.9029  | 0.0971   |
| 373.47      | 50040    | 0.6144  | 0.3856   | 0.9251  | 0.0749   |
| 373.47      | 58680    | 0.7769  | 0.2231   | 0.9560  | 0.0440   |
| 373.47      | 66370    | 0.8617  | 0.1383   | 0.9746  | 0.0254   |
| 373.47      | 75380    | 1.0000  | 0.0000   | 1.0000  | 0.0000   |

1. Determine whether that experimental data are thermodynamically consistent.
2. Determine the functional expressions for liquid activity coefficients using the Van Laar model for the **Gibbs Excess** energy function.
3. Estimate from the experimental data the Van Laar model coefficients  $A_{12}$  and  $A_{21}$  and draw a parity plot of predicted pressure values vs. experimental values.

## BUBBLE POINT TEMPERATURE with an EOS

Step 1. Set the Liquid Composition ( $x$ ) equal to the overall composition ( $z$ )  $\Rightarrow x=y$   
Copy  $\rightarrow$  Paste Special  $\rightarrow$  Values

Step 2. Guess a vapor composition ( $y$ )  $\Rightarrow y^0 = x$   
Copy  $\rightarrow$  Paste Special  $\rightarrow$  Values

Step 3. Calculate the compressibility factor  $Z_v$  for vapor and  $Z_L$  for liquid using a Temperature guess ( $T^0$ ) usually the experimental value (if any)  
If  $Z_v > Z_L$  proceed to step 4. Otherwise guess another temperature

Step 4. Calculate the fugacity coefficients  $\hat{\phi}_i^v$  and  $\hat{\phi}_i^L$  for the vapor and liquid phases

Step 5. Calculate the K-values  $K_i = \frac{\hat{\phi}_i^L}{\hat{\phi}_i^v}$  and in the Bubble Point Temperature menu group click Solve Bubble Point Equation to adjust the temperature value

Step 6. Recalculate the vapor composition  $y = Kx$  using the newly calculated K-values

### ITERATION STEP

We are now ready to start the iteration for the calculation of the bubble point temperature. In the Bubble Point Temperature menu group we click Update Y-values until the New Molar composition values for the vapor phase become the same as the old vapor composition values. Usually that will happen in 2-3 iterations.

### DEW POINT TEMPERATURE with an EOS

The Dew point temperature calculation follows the same steps outlined in the Bubble point temperature calculation; but now we set the vapor composition ( $y$ ) equal to the overall composition ( $z$ )  $\Rightarrow y = z$ , and in the iteration step we now update the  $x$ -values, until convergence, while we are solving the Dew Point Equation.

### BUBBLE & DEW POINT PRESSURE with an EOS

Bubble and Dew point pressure are calculated using exactly the same steps outlined below, but now solving the bubble and dew point pressure equation

### Question #1 – VLE Calculation with Equations of State

**Part 1** – Assuming BIP is zero, plot the bubble point temperature vs liquid mole fraction and dew point temperature vs vapor mole fraction.

$$k = \frac{\phi_L}{\phi_V} \quad (1)$$

$$k = \frac{y_i}{x_i} \quad (2)$$

For this part of the problem, we are given the liquid and vapor mole fractions of the two components in the system that were measured at their given temperatures. We are also told that at one of the temperatures, there is an azeotrope and which temperature that is.

The first part of the problem is to plot the bubble point temperature vs the liquid mole fraction. We already have the liquid mole fraction, so all we need to do is calculate the bubble point temperature. The bubble point temperature is defined as the temperature where the very first bubble starts to form. In this case the mixture will be mostly liquid, so we can assume that the overall composition is the same as the liquid composition that we are given. We don't know what the composition of the vapor is yet, but once more we can assume them to be the liquid compositions as a starting point.

The first thing to do would be to use the CHE301 computational tool and calculate the compressibility factor for both the liquid and the vapor guessing the experimental temperature as the initial temperature and an initial compressibility factor of 1 and 0 for the vapor and liquid respectively. This way, once we have a compressibility factor that we can work with, we can find the fugacity coefficient for each compound in their phases. The reason we want to do this is because with the fugacity coefficient of the compounds in the mixture, we can calculate the k value using Equation (1).

The k value is used to relate liquid composition and vapor composition, but since initially the vapor composition was unknown, we can instead use the fugacity coefficients and use the k values found from the coefficients to find new molar compositions for the vapor. Once everything is set and ready to go, the first step would be to solve for the bubble point by setting the bubble point equation equal to 0 and recurring through each new y value determined from Equation (2) until the new vapor composition is the same as the old vapor composition. At the liquid composition given, we can now calculate the bubble point temperature for each mole fraction (Table I).

**Table I.** Experimental Liquid Fractions and Corresponding Bubble Point Temperatures determined using the Peng Robinson Equation of State

| x <sub>l</sub> | T <sub>bubble</sub> |
|----------------|---------------------|
| 0.0000         | 353.1905            |
| 0.0260         | 352.0170            |
| 0.0500         | 351.0045            |
| 0.0880         | 349.5271            |
| 0.1640         | 346.9727            |
| 0.3330         | 342.7654            |
| <b>0.6140</b>  | <b>338.7709</b>     |
| 0.5490         | 339.4427            |
| 0.6990         | 338.0775            |
| 0.7820         | 337.5866            |
| 0.8980         | 337.2223            |
| 0.9730         | 337.2588            |
| 1.0000         | 337.3538            |

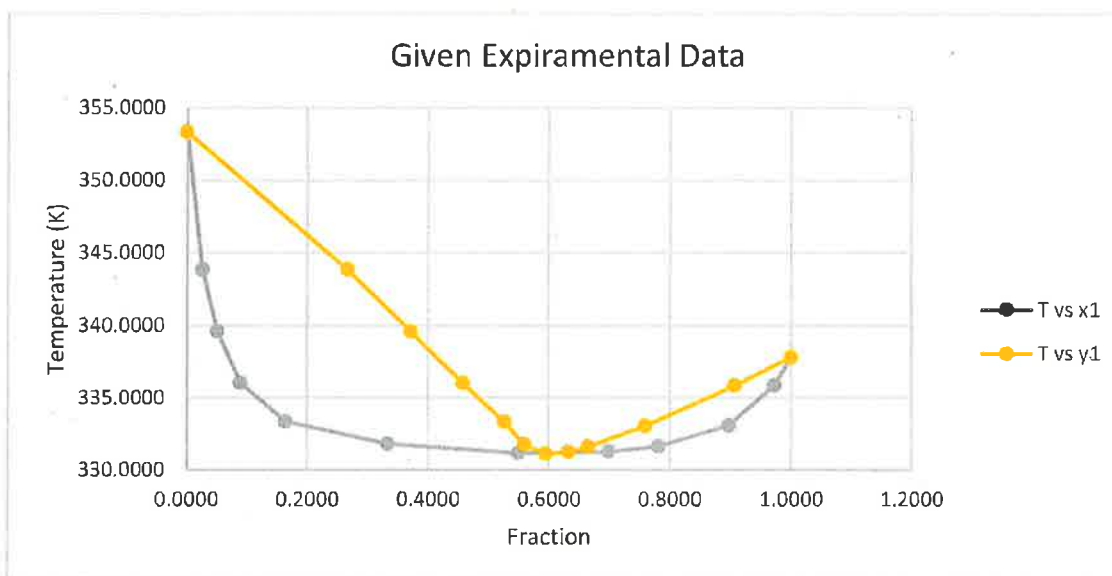
The second part of the problem is like the first part, but it asks us to find the dew point temperature and plot it vs the vapor mole fraction. Although, since the dew point is defined as the temperature at which the first drop of liquid is created, we will be working with the given vapor compositions and not the liquid compositions to solve for the temperature. Similarly, to the previous case, here we know our mixture is mostly vapor so we can assume that the overall composition is the same as the vapor composition and we will be using the vapor composition themselves simply as an initial guess for the liquid composition.

Once more, we will use the CHE301 computational tool to calculate the compressibility factor, the fugacity coefficient, and the  $k$  values for each compound using the same methodology we did previously. This time, the  $k$  value will be used to find the liquid composition. Once the dew point equation is set to 0 by changing the temperature, a recurrence through each  $x$  value until the new values determined from  $k$  and the old values that were the initial guesses are the same will determine the dew point temperature (Table II).

**Table II.** Experimental Vapor Fractions and Corresponding Dew Point Temperatures determined using the Peng Robinson Equation of State

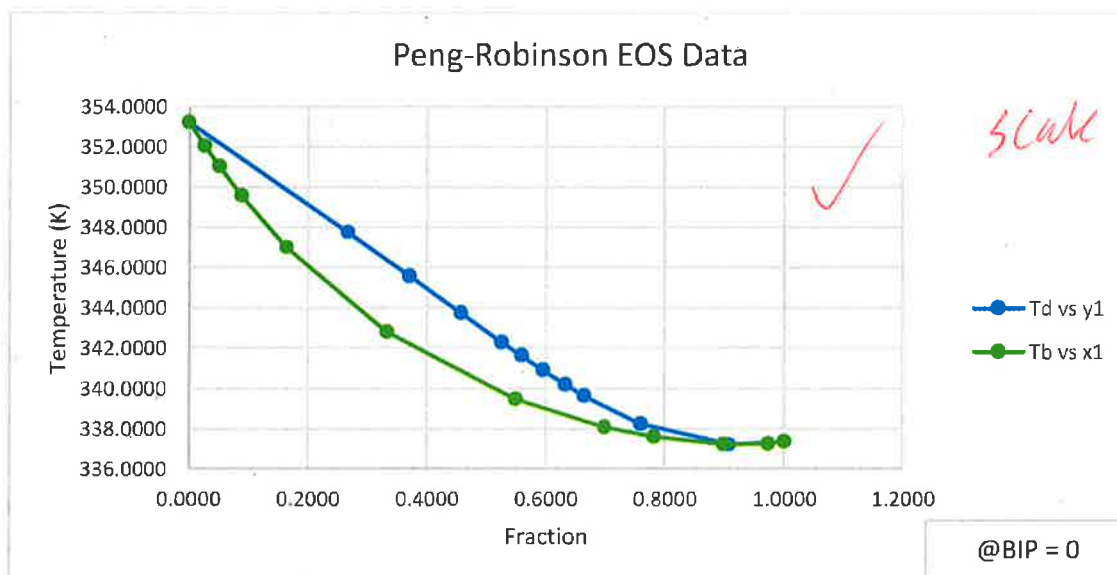
| $y_1$         | T <sub>dew</sub> |
|---------------|------------------|
| 0.0000        | 353.1905         |
| 0.2670        | 347.7315         |
| 0.3710        | 345.5285         |
| 0.4570        | 343.7074         |
| 0.5260        | 342.2741         |
| 0.5590        | 341.6073         |
| <b>0.6140</b> | <b>340.5401</b>  |
| 0.5950        | 340.9012         |
| 0.6330        | 340.1886         |
| 0.6650        | 339.6228         |
| 0.7600        | 338.2128         |
| 0.9070        | 337.2169         |
| 1.0000        | 337.3538         |

The original data, when plotted provides us with the following relationship between the experimental temperatures and the vapor/liquid molar compositions that were determined.



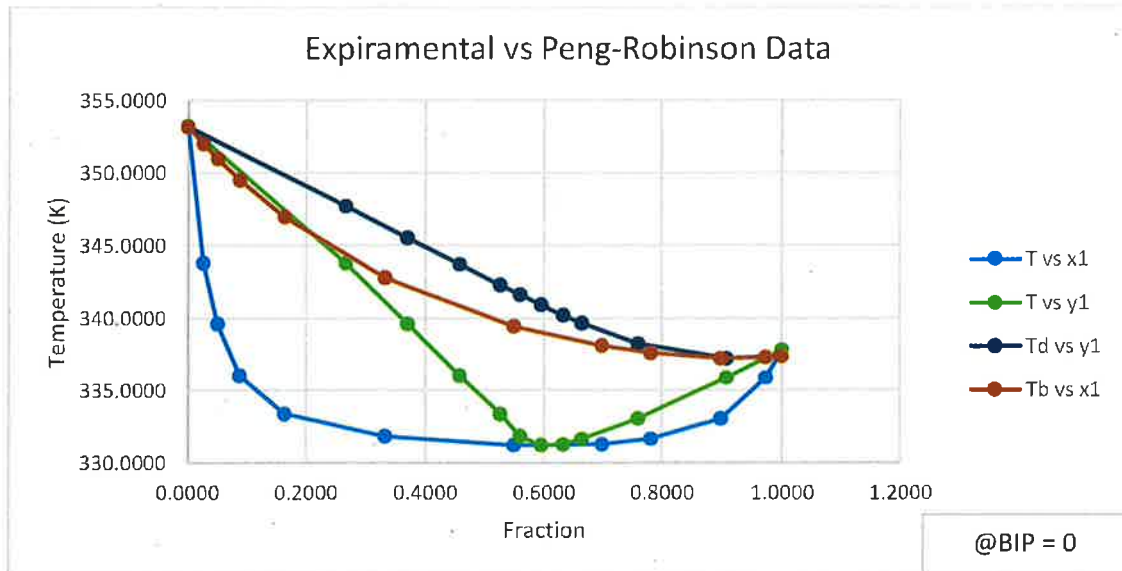
**Figure 1.1** The figure above denotes the relationship between the experimental temperatures and the measured experimental liquid compositions for component 1 (yellow, top). It also denotes the relationship between the experimental temperatures and the measured experimental vapor compositions for component 1 (gray, bottom).

Our calculated data can also be plotted to provide us with the relationship between the bubble and dew point temperatures at the vapor/liquid compositions.



**Figure 1.2** The figure above denotes the relationship between the calculated Bubble temperatures and the measured experimental liquid compositions for component 1 (blue, top). It also denotes the relationship between the calculated Dew temperatures and the measured experimental vapor compositions for component 1 (green, bottom). The binary interaction parameter is assumed to be zero. Data taken from Table I and Table II.

Taking the data determined from the iteration done previously under the assumption of the Peng-Robinson equation of state, we can superimpose our graph relationship unto the true relationship to see how accurate that equation of state really is.

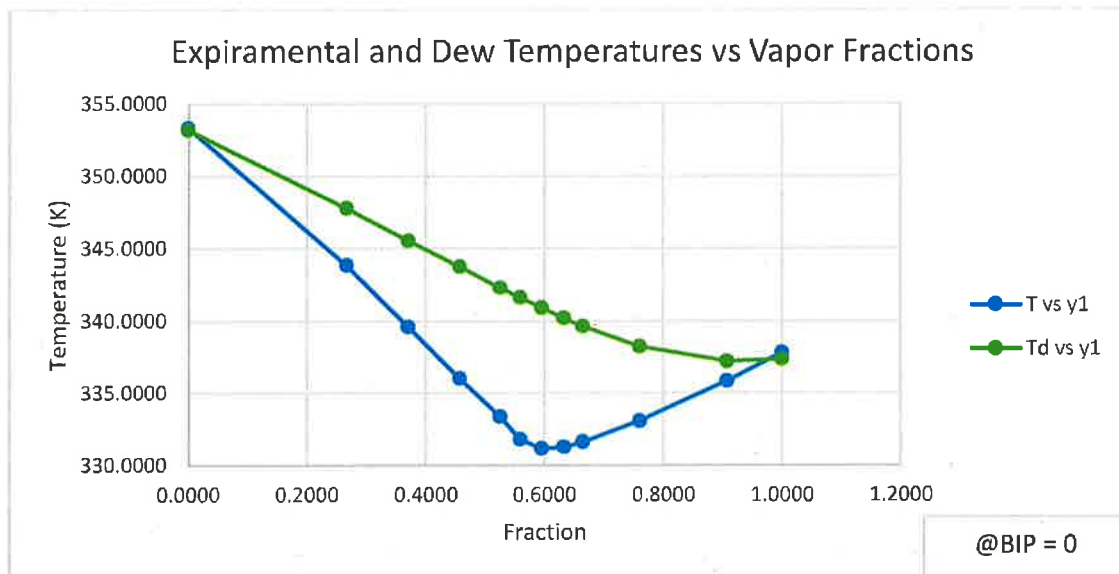


**Figure 1.3** The figure above combined previous figures (1.1) and (1.2) all under one graph to demonstrate the efficiency (or lack of) when using the Pen-Robinson equation of state in an attempt to model the mixture of methanol and benzene. The binary interaction parameter is assumed to be zero.

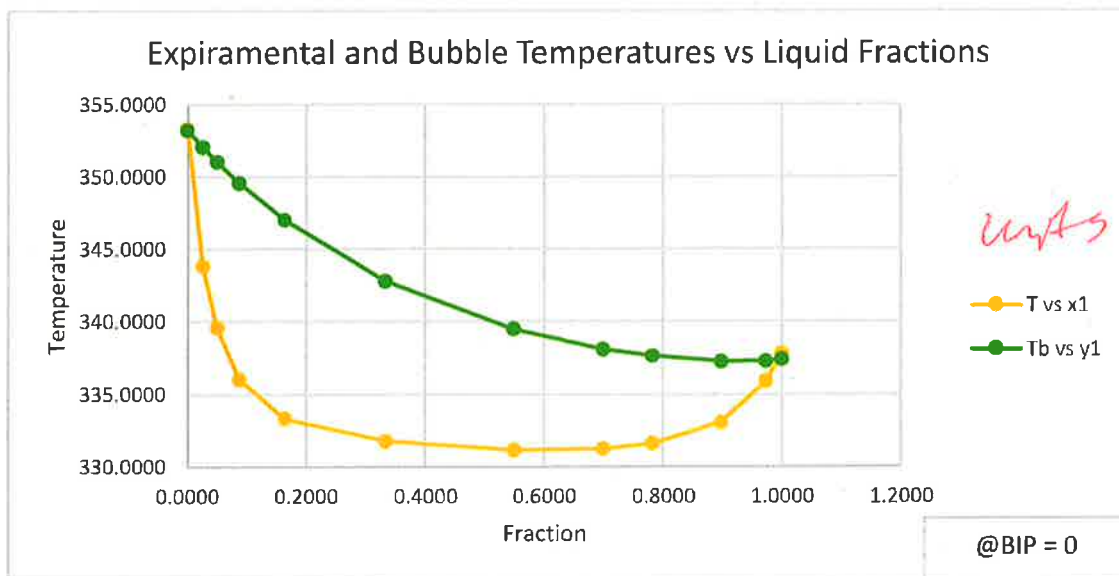
The visual fit is horrendous. The correlation between the two graphs is almost non-existent and the calculated data cannot be used as an accurate representation of our determined experimental data. Numerically, we can determine how well the calculated data truly represents the experimental data by finding out the mean residual sum of squares. We do this by subtracting the experimental temperature from the bubble and likewise for the dew and then squaring.

Taking the average of those two values gives us a MRSS of 776.49. If this value were below 10, then we could say that the calculated values using the PR EOS is acceptable.

We can also compare two graphs where we know they should be the same but aren't take a deeper look at how dysfunctional the PR EOS fit is for our system.



**Figure 1.4** The figure above denotes the relationship between the calculated Dew temperatures and the measured experimental vapor compositions for component 1 (green, top). It also denotes the relationship between the experimental temperatures and the measured experimental vapor compositions for component 1 (blue, bottom). The binary interaction parameter is assumed to be zero. Data taken from Table II.



**Figure 1.5** The figure above denotes the relationship between the calculated Bubble temperatures and the measured experimental liquid compositions for component 1 (green, top). It also denotes the relationship between the experimental temperatures and the measured experimental liquid compositions for component 1 (yellow, bottom). The binary interaction parameter is assumed to be zero. Data taken from Table I.

**Part 2** – Use trial and error to determine the temperature and composition predicted for the azeotrope between methanol and benzene if the BIP is zero.

Looking at the graph that we generated, our azeotrope is located farther down the line at a composition of about 0.9 and a much higher temperature. A simple eyeball guess could get us to predict an initial value for where the azeotrope might be. Like what we did in part one, we can set the liquid, overall, and vapor composition equal to our azeotropic compositions determined from the PR EOS. We can use the relationship of Equation (1) and Equation (2) to find our values for the new vapor compositions and liquid compositions of methanol and benzene at the azeotrope. The old and the new compositions in both the liquid and the vapor should be the same but also the compositions in the liquid and vapor should be equal of an azeotrope. Initially, playing around with trial and error gets us in the ball park of which temperature would satisfy the azeotrope conditions with a BIP of zero.

**Table III.** Reported Azeotropic compositions based off Peng-Robinson Equation of State approximation and Azeotropic occurrence temperature.

|                             | Composition |        |
|-----------------------------|-------------|--------|
|                             | $x_i$       | $y_i$  |
| <b>Azeotrope</b>            | 0.9         | 0.1    |
| <b>Methanol</b>             | 0.8996      | 0.0948 |
| <b>Benzene</b>              | 0.1055      | 0.0948 |
| <b>Temperature<br/>(°C)</b> | 63.9500     |        |

The individual compositions couldn't be perfectly matched up but a close enough value could be determined. The biggest issue is that the azeotrope temperature was found to be about 64°C, which we know to be very far off from the temperature determined from experiments (58°C). The compositions are of the same nature where our determined azeotropic composition is much higher than it should be.

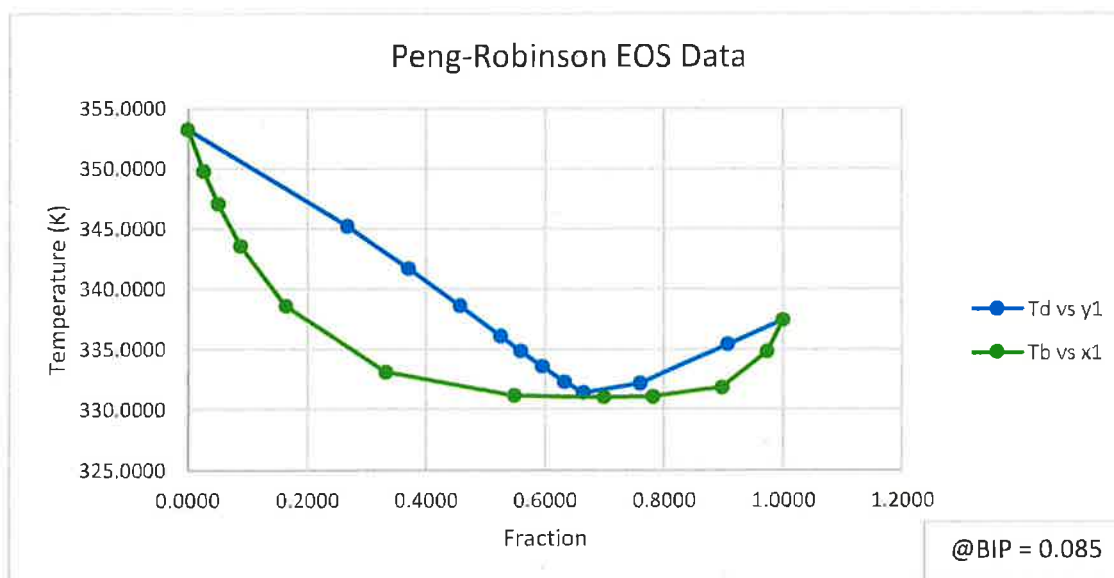
**Part 3** – Determine the optimal value of the BIP that can be used for the prediction of the azeotrope of this mixture

For this part of the problem, we already know that the PR EOS doesn't fit well but if the binary interaction parameter could be somehow changed from zero to something else to give it a better fit, the goal would be what that parameter is.

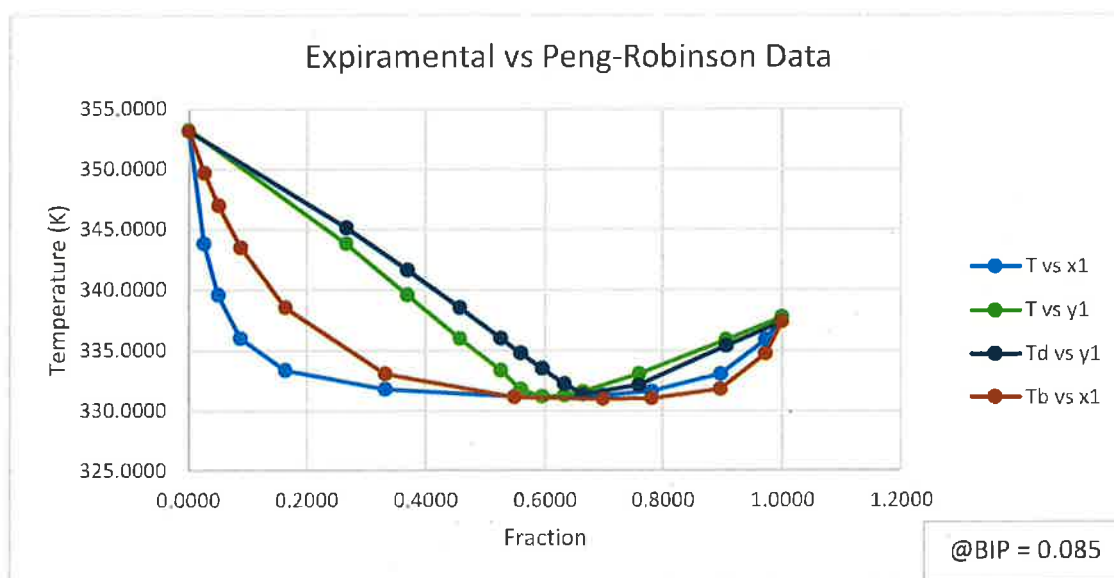
For our binary interaction parameter to be optimal, our azeotrope temperatures should be the same as well as our compositions. In the excel document, we could go back and replace the values we eyeballed for our azeotrope with the real azeotrope, we can set up about 5 cells to be the squared residuals the temperature where our azeotrope is at and the experimental azeotrope and the residual of our y and x values. We can use solver to change our temperature and our BIP where the squared residuals are constrained to a value of zero. Solver won't be able to solve it but it gives us a value that is close to what we need and keep the residuals at a minimum.

**The BIP ends up equaling about 0.085 and our Temperature converged to 57.96°C.**

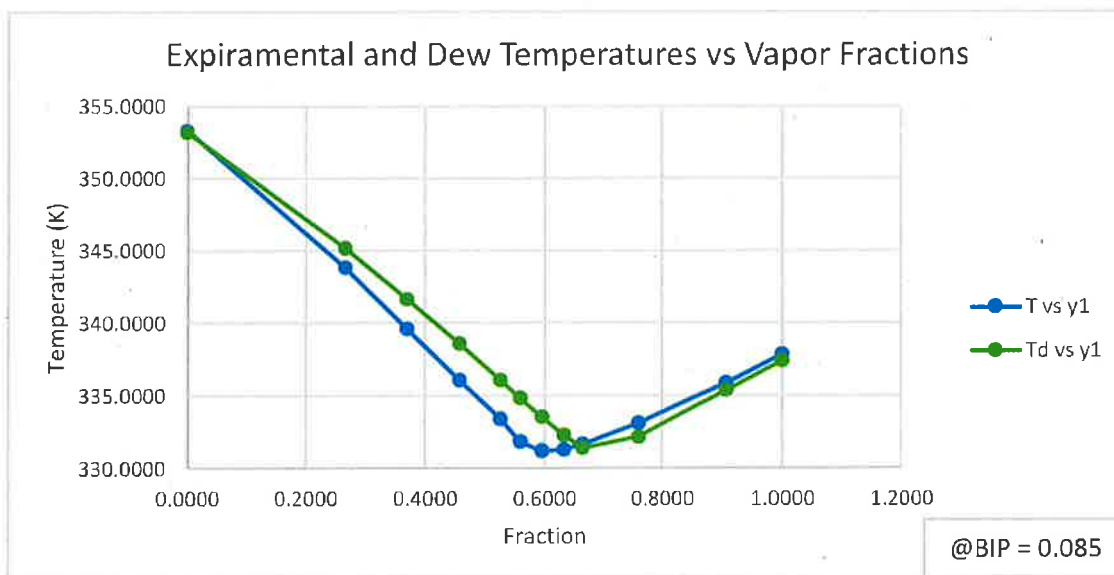
In theory, this new value for the BIP should be the near perfect addition to the PR EOS to adjust the graphs and make our model refined. If we go back and repeat the entire procedure of part 1 and acquire our graphs, it is obvious that not only is the BIP correct in fixing the PR EOS but also does a really good job doing it.



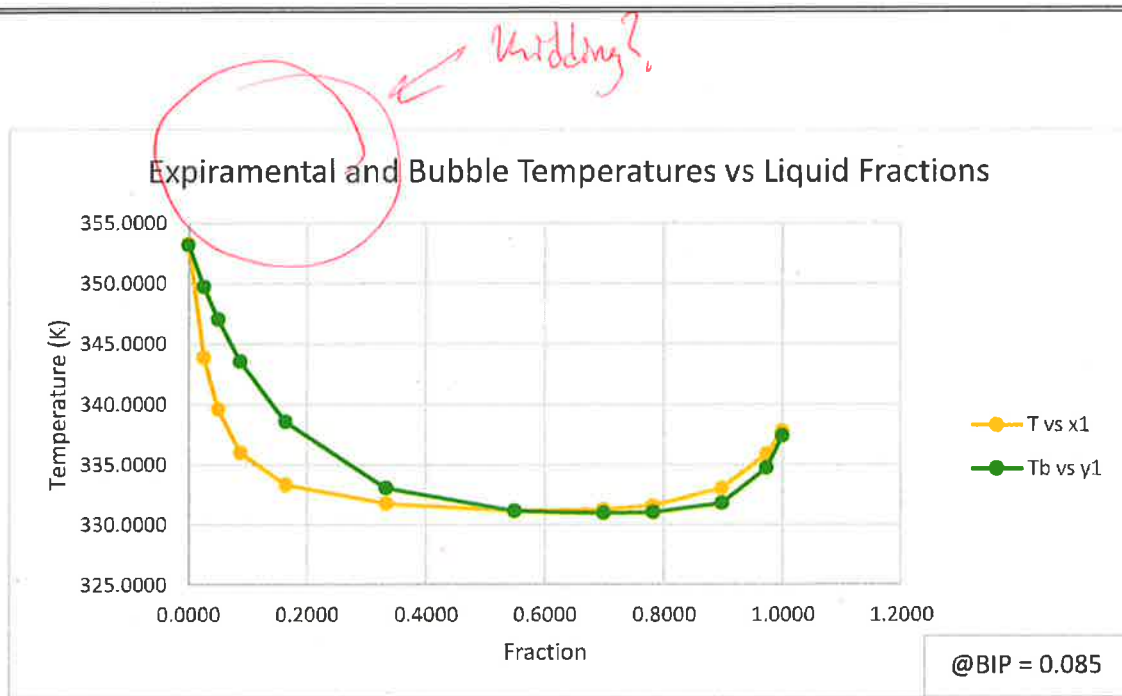
**Figure 1.6** The figure above denotes the relationship between the calculated Bubble temperatures and the measured experimental liquid compositions for component 1 (blue, top). It also denotes the relationship between the calculated Dew temperatures and the measured experimental vapor compositions for component 1 (green, bottom). The optimal binary interaction parameter is determined to be 0.085. This graph is related to figure (1.2).



**Figure 1.7** The figure above combined previous figures (1.8) and (1.9) all under one graph to demonstrate the efficiency (or lack of) when using the Pen-Robinson equation of state in an attempt to model the mixture of methanol and benzene. The optimal binary interaction parameter is determined to be 0.085. This graph is related to figure (1.3).



**Figure 1.8** The figure above denotes the relationship between the calculated Dew temperatures and the measured experimental vapor compositions for component 1 (green, top). It also denotes the relationship between the experimental temperatures and the measured experimental vapor compositions for component 1 (blue, bottom). The optimal binary interaction parameter is determined to be 0.085. This graph is related to figure (1.4).



**Figure 1.9** The figure above denotes the relationship between the calculated Bubble temperatures and the measured experimental liquid compositions for component 1 (green, top). It also denotes the relationship between the experimental temperatures and the measured experimental liquid compositions for component 1 (yellow, bottom). The optimal binary interaction parameter is determined to be 0.085. This graph is related to figure (1.5).

## Question #2 – VLE Calculations with Liquid Activity Coefficients

### Part 1 – Determine whether the experimental data are thermodynamically consistent

In this problem, we are given the experimental data and are asked to use the Van Laar model to find the liquid activity coefficients at each varying pressure. The thermodynamic consistency is used to determine whether the experimental data provided is questionable or reliable. The threshold for it is  $\pm 0.1$  and can be modeled by the following formula.

$$\ln(\gamma_1) = \frac{G^E}{RT} = \frac{x_1 x_2 A_{12} A_{21}}{x_1 A_{21} + x_2 A_{12}} \quad (1)$$

$$\ln\left(\frac{\gamma_1}{\gamma_2}\right) - \ln\left(\frac{\gamma_1^{EXP}}{\gamma_2^{EXP}}\right) = TC \quad (2)$$

$$\gamma_i^{EXP} = \frac{y_i^{EXP}}{x_i^{EXP}} * \frac{P}{P_i^{sat}} \quad (3)$$

To solve for the thermodynamic consistency, we would first see what happens if the component we are working with was a pure component rather than a mixture. Looking at equation (1), if either the 2<sup>nd</sup> component or the first component were zero making the other a pure substance, then the log of the activity coefficient would be zero. That means we can apply the Van Laar model.

Next, we can plot  $\ln\left(\frac{\gamma_1^{EXP}}{\gamma_2^{EXP}}\right)$  vs the experimental  $x_1$  data. If we are thermodynamically consistent, then we can fit our plot with best fit linear line, integrate from 0 to 1, and if that value is within our threshold then we can say that we are thermodynamically consistent.

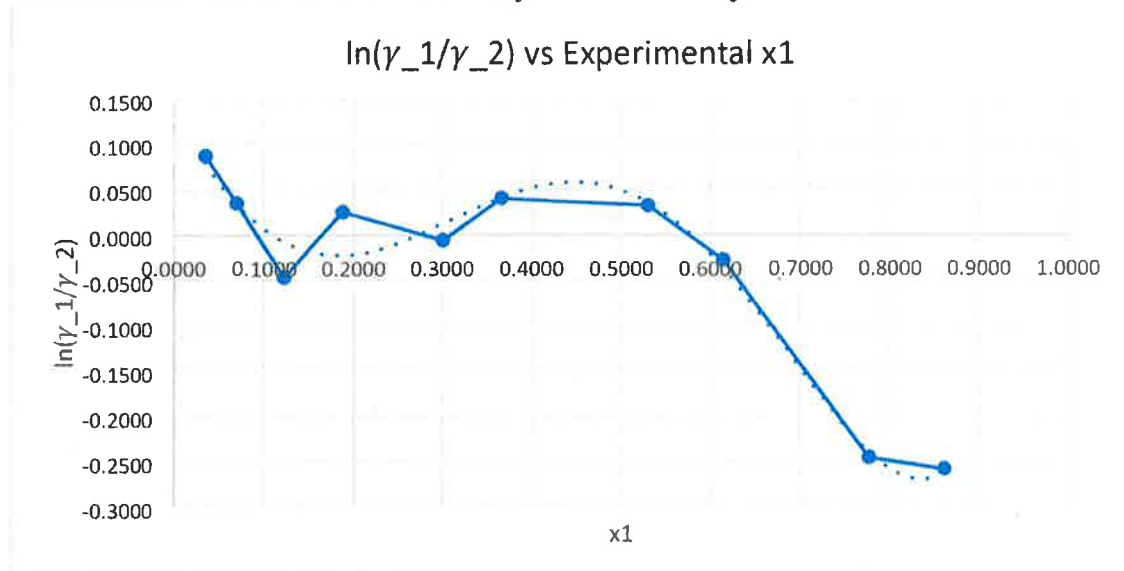
**Table IV.** Experimental log of activity coefficient ratios determined from 2<sup>nd</sup> term of equation (2) as well as experimentally determined liquid molar fractions.

| THERMO DYNAMIC CONSISTENCY |                          |
|----------------------------|--------------------------|
| $x_1$                      | $\ln(\gamma_1/\gamma_2)$ |
| 0.0353                     | 0.0869                   |
| 0.0700                     | 0.0350                   |
| 0.1223                     | -0.0469                  |
| 0.1880                     | 0.0250                   |
| 0.3006                     | -0.0056                  |
| 0.3665                     | 0.0401                   |
| 0.5301                     | 0.0328                   |
| 0.6140                     | -0.0269                  |
| 0.7769                     | -0.2457                  |
| 0.8617                     | -0.2587                  |
| 0.0353                     | 0.0869                   |
| 0.0700                     | 0.0350                   |

From table IV, we can generate a plot from which we can create a polynomial function for our data. Different order polynomial fits provide us with different values for the total area. A polynomial of order 2 provides the best value for the thermodynamic consistency but doesn't do a good job at representing the plotted points. A polynomial of order 5 on the other hand does both a good job at fitting our experimental data and providing a value for the thermodynamic consistency  $< |0.1|$ .

*No polynomial function*  
-10.

The area beneath the curve for the thermodynamic consistency is -0.012.



**Figure 2.1** The figure above denotes the relationship between or natural log of the activity coefficients ratios and the provided experimental data determined for the liquid fraction. The polynomial fit is  $13.62x^5 - 17.67x^4 - 0.43x^3 + 6.53x^2 - 1.99x + 0.1417$ . This figure uses the data from Table IV to determine the thermodynamic consistency of the data provided.

**Part 2** – Determine the functional expressions for the LAC using Van Laar model for Gibbs Excess energy function

The Van Laar model for Gibbs Excess which has 2 parameters and the activity coefficient can be modeled by the following equation:

$$\ln \gamma_1 = \frac{G^E}{RT} - x_2 \frac{d}{dx_2} \left( \frac{G^E}{RT} \right) \quad (4)$$

Before taking the derivative of the Van Laar, we need to make a substitution for  $x_1$  since it's a function of  $x_2$ . The addition of these two mole fractions is simply one so the form of the equation for taking a derivative becomes:

$$\frac{G^E}{RT} = \frac{x_2}{x_2(A_{12} - A_{21}) + A_{21}} + \frac{x_2^2}{x_2(A_{12} - A_{21}) + A_{21}} \quad (5)$$

All the "A" value are constants, so we take the derivative of both terms and combine, this is the result:

$$\frac{d \left( \frac{G^E}{RT} \right)}{dx_2} = \frac{-(A_{12} - A_{21})x_2^2 - 2A_{21}x_2 + A_{21}}{[x_2(A_{12} - A_{21}) + A_{21}]} \quad (6)$$

Equation (1) and Equation (6) can now be plugged into equation (4) to give us the overall value for the liquid activity coefficient for compounds 1. Compound 2's activity coefficient can just as easily be determined simply from inverting the vertices.

$$\ln \gamma_1 = \frac{x_1 x_2 A_{12} A_{21}}{x_1 A_{21} + x_2 A_{12}} - A_{21} A_{12} x_2 \left( \frac{-(A_{12} - A_{21})x_2^2 - 2A_{21}x_2 + A_{21}}{[x_2(A_{12} - A_{21}) + A_{21}]} \right) \quad (7)$$

$$\ln \gamma_2 = \frac{x_2 x_1 A_{21} A_{12}}{x_2 A_{12} + x_1 A_{21}} - A_{12} A_{21} x_1 \left( \frac{-(A_{21} - A_{12})x_1^2 - 2A_{12}x_1 + A_{12}}{[x_1(A_{21} - A_{12}) + A_{12}]} \right) \quad (8)$$

Equation (7) and Equation (8) were now programmed into excel via VBA and the respected gamma values were found for the different molar compositions at a constant temperature and a varying pressure. We now have a way to find the liquid activity coefficients at the assumed value for the constants and as well as the first term from Equation (2).

*This can be simplified*

*-8*

**Part 3**– Estimate from the experimental data the Van Laar model coefficients and draw a parity plot of predicted pressure values vs. experimental values.

The pressure can be predicted using the 2 gamma values as well as the 2 liquid molar compositions with the saturated vapor pressures of the 2 compounds using equation (8).

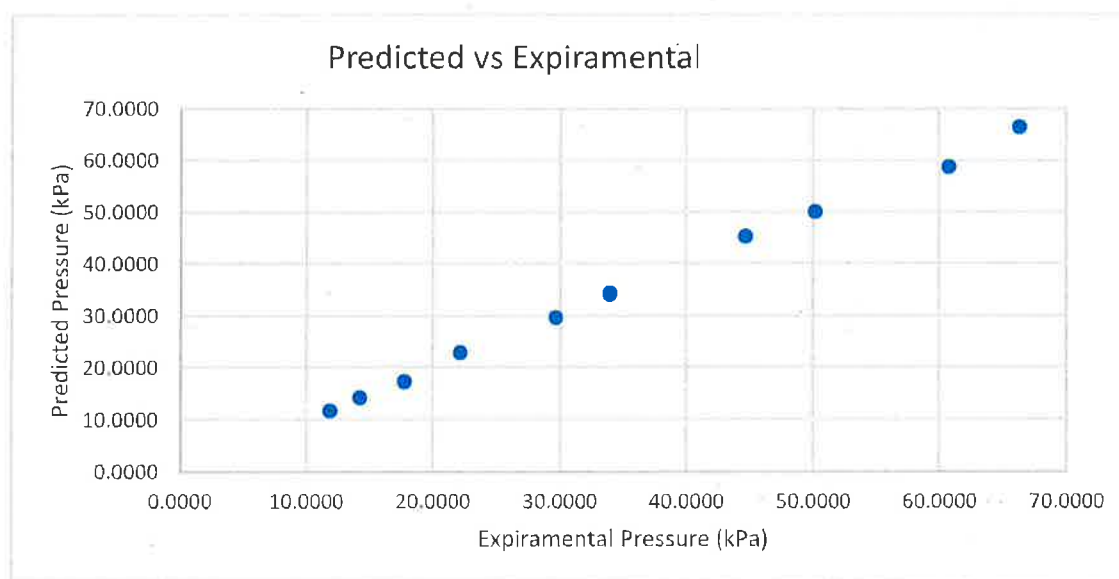
$$P_{predicted} = x_1 \gamma_1 P_1^{sat} + x_2 \gamma_2 P_2^{sat} \quad (9)$$

The overall goal is to estimate the  $A_{21}$  and  $A_{12}$  coefficients. To do this, we find the residuals of our pressure by subtracting the predicted from the true pressure and summing the squares. Using the solver functionality in excel to find a minimum value for the sum of the squared residuals while changing the  $A_{12}$  and the  $A_{21}$  values. This way the predicted temperatures would be as close as possible to the experimental temperatures. We can also plot our predicted pressure vs the experimental pressure in the form of a parity plot to see how good the data really is visually.

**Our optimal  $A_{12}$  and  $A_{21}$  values are 0.0255 and 0.0429 respectively.**

**Table V.** Experimental Temperatures at which the data was recorded, and the predicted temperatures calculated from Equation (9)

| Predicted | Experimental |
|-----------|--------------|
| 11.8814   | 11.6300      |
| 14.2493   | 14.1800      |
| 17.7841   | 17.2400      |
| 22.1785   | 22.7100      |
| 29.6278   | 29.5300      |
| 33.9563   | 34.2600      |
| 44.6526   | 45.2400      |
| 50.1275   | 50.0400      |
| 60.7639   | 58.6800      |
| 66.3110   | 66.3700      |



**Figure 2.** The figure above is a parity plot of the predicted pressures and the experimental pressures. The parity plot is fitted with a linear model and the equation as well as the  $R^2$  is provided in the top right corner. The data for this plot is taken from Table V